Exact popularity probabilities in house allocation

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Abstract

We classify exact popularity probabilities for a fixed house allocation under three representations of uncertain preferences. Independent lotteries and independent uniform tie refinements give $\mathrm{FP}^{\#\mathrm{P}}$-complete evaluation and PP-complete threshold comparison under polynomial-time Turing reductions. Explicit joint lotteries admit polynomial-time evaluation. The results allow incomplete acceptable lists and partial matchings.

Applicants have strict preferences over their acceptable houses; every acceptable house is better than being unmatched. Houses have no preferences. A matching is *popular* if no other feasible matching receives strictly more votes in their pairwise comparison. The acceptable pairs are fixed across all realized profiles. For an encoded distribution I and a feasible fixed matching M, write

$$\pi_I(M) = \Pr(M \text{ is popular}).$$

This is the probability-evaluation question in catalogue entry C78-387, motivated by the uncertainty agenda of Cechlárová, Cseh and Manlove [1, §3.1.2].

Theorem 1. *For independent explicitly listed rational lotteries over applicants' orders, and for independent uniform refinements of weak orders, exact rational evaluation of $\pi_I(M)$ is $\mathrm{FP}^{\#\mathrm{P}}$-complete. Deciding $\pi_I(M) \geq \tau$, for a binary rational $\tau \in [0,1]$, is* PP*-complete. Every completeness statement uses polynomial-time Turing reductions. For an explicitly listed rational lottery over whole profiles, both tasks are in* P*.*

Verification and upper bounds. For a realized profile, weight an acceptable pair (a,h) by $+1$, 0 or -1 according as a prefers h to $M(a)$, receives the same house, or prefers $M(a)$ to h. Give each applicant a private dummy for being unmatched, with weight 0 if unmatched in M and -1 otherwise. A maximum-weight assignment to real or private dummy houses computes the largest vote margin against M. Since M has margin zero, it is popular exactly when this maximum is zero. Weighted bipartite matching computes this in polynomial time, including when M is partial.

For an independent lottery, use a common denominator D_a for applicant a's listed probabilities. Integer tickets in $[0, D_a)$ select its order by cumulative intervals. With $D = \prod_a D_a$, let F count successful ticket tuples; then $\pi_I(M) = F/D$. For uniform refinements take $D = \prod_C |C|!$, over all tie blocks C, and decode each block ticket as a permutation. In both cases D has polynomial bit length and F is a #P function: guess uniquely encoded tickets, reject invalid encodings, and test popularity. Exact evaluation therefore lies in $\mathrm{FP}^{\#\mathrm{P}}$.

Number the ticket tuples by $0, \ldots, D-1$ using mixed-radix decoding. Here is an explicit PP test, including equality. Write $\tau = u/v$, where $0 \leq u \leq v$ and $v > 0$, and choose K with $2^K > vD$. Guess one bit b and a K-bit integer t. If $b = 0$, accept $t = vD$; for $t < vD$, accept exactly when profile ticket $\lfloor t/v \rfloor$ succeeds; reject the rest. If $b = 1$, accept exactly when $t \geq uD$. The number of accepting branches is

$$(vF + 1) + (2^K - uD),$$

which exceeds half of the 2^{K+1} branches exactly when $vF \geq uD$. For a listed joint lottery, test every support profile and sum its rational weight; this takes polynomial time.

A vote lemma. In a balanced instance label the assigned houses h_a, so that the fixed matching is $M(a) = h_a$. Call a *top* if h_a is its first acceptable house.

Lemma 1. *The perfect matching M is popular if and only if every house preferred to an applicant's assigned house is assigned in M to a top applicant.*

Proof. Under the stated condition, a winner against M takes a top applicant's house. Its owner must lose that house, and therefore votes for M. Distinct winners take distinct houses, giving an injection from winners to losers.

Conversely, suppose a prefers h_b to h_a and b is not top. Choose h_c preferred by b to h_b. If $c = a$, swapping a and b gives two wins and no losses. Otherwise give a house h_b, give b house h_c, and leave c unmatched. There are two wins and one loss. All newly assigned pairs are acceptable, so either alternative defeats M. □

Independent lotteries. Given a simple graph $G = (V, E)$, create applicant a_v and house h_v for each vertex, with $M(a_v) = h_v$. Every house is acceptable. At each nonisolated vertex choose independently, with equal probabilities, between putting its own house first and putting all neighbors before it. All nonneighbors other than its own house follow it; fix strict orders within the neighbor and nonneighbor groups. Isolated vertices have a deterministic order with their own house first.

Let S consist of the vertices choosing the second order. If S is independent, all improved houses belong to top applicants, so Lemma 1 applies. If $u, v \in S$ are adjacent, their house swap gives two wins. Thus popularity is equivalent to independence. If G has z isolates, the number of choices is $2^{|V|-z}$ and the number of independent sets is $2^z i(G - \text{isolates})$. Hence

$$\pi_I(M) = \frac{i(G)}{2^{|V|}}. \tag{1}$$

Provan and Ball establish #P-completeness for counting bipartite independent sets [2, §1, problem 2; §2, reduction 2]. Equation (1) proves the required counting hardness.

Uniform tie refinements. For each vertex v make an applicant a_v with the sole tie $\{h_v, g_v\}$, and an applicant z_v whose only acceptable house is g_v. Match them to h_v and g_v, respectively. For each edge uv make applicants $e_{uv,j}$, $1 \le j \le k$, each with sole tie $\{h_u, h_v, h_{uv,j}\}$ and assigned private house $h_{uv,j}$. All refinements are independent and uniform.

Let $X_v = 1$ when a_v ranks h_v above g_v. By Lemma 1, popularity holds exactly when every edge applicant puts its own house above each endpoint house h_v with $X_v = 0$. Among the six strict edge orders, this has conditional probability

$$w(0,0) = \tfrac{1}{3}, \qquad w(0,1) = w(1,0) = \tfrac{1}{2}, \qquad w(1,1) = 1.$$

The resulting probability is

$$\pi_k = 2^{-|V|} \sum_{X \in \{0,1\}^V} \prod_{uv \in E} w(X_u, X_v)^k. \tag{2}$$

Put $m = |E|$. For $S = \{v : X_v = 0\}$ let a count edges inside S and b edges inside its complement. If N_{ab} counts such subsets, then

$$2^{|V|+km}\pi_k = \sum_{\substack{a,b \ge 0 \\ a+b \le m}} N_{ab} \left(\frac{2^{a+b}}{3^a} \right)^k. \tag{3}$$

The $t = (m+1)(m+2)/2$ positive rational bases are distinct: their powers of 3 determine a, then their powers of 2 determine b. Values at $k = 1, \ldots, t$ form a nonsingular Vandermonde system and recover every N_{ab}, hence

$$i(G) = \sum_{b=0}^{m} N_{0b}.$$

There are $O(m^2)$ queries, the largest has $O(|V| + m^3)$ applicants, and the matrix entries have $O(m^3)$ bits. Exact rational elimination therefore takes polynomial time. When $m = 0$, return $2^{|V|}$ directly. This establishes counting hardness using ties of size at most three.

Completeness. Either evaluation oracle computes the #P-complete independent-set count, so it can replace every #P oracle call. This gives $\mathrm{FP}^{\#\mathrm{P}}$ hardness. Conversely, a threshold oracle recovers the successful-ticket count $F \in [0, D]$ by binary search at thresholds j/D, using $O(\log(D+1))$ queries. It therefore simulates either exact-evaluation oracle. Every PP language is decidable by a #P count followed by comparison with half its computation paths, giving PP hardness under the stated Turing reductions. The earlier upper bounds finish Theorem 1.

Attribution. The independent-set construction and tie interpolation are already present in the catalogue's Claude Opus 5.5 High attempt; its GPT 6 Astra Ultra attempt gives the general verification and counting upper bounds. We combine these arguments, repair the isolated-vertex count, supply direct vote proofs and exact bit bounds, and complete the threshold classification. No many-one completeness claim is made.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] K. Cechlárová, Á. Cseh and D. F. Manlove. *Selected open problems in Matching Under Preferences* (2019), §3.1.2. `https://hpi.de/friedrich/docs/publications/2019/BEATCS.pdf`.

[2] J. S. Provan and M. O. Ball. The complexity of counting cuts and of computing the probability that a graph is connected. *SIAM Journal on Computing* **12** (1983), 777–788. `https://doi.org/10.1137/0212053`.